

Model Card: Solar Rooftop PV Detection Model

TEAM: NEURAL NOMADS

TEAM MEMBERS:

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1. Model Overview

Model Name: Solar Rooftop PV Detection Model

Version: v1.0

Task: Rooftop solar photovoltaic (PV) detection and segmentation

Model Type: Deep learning segmentation model

Architecture: YOLOv8 Segmentation

This model is designed to detect rooftop-mounted solar photovoltaic panels from high-resolution satellite imagery using geographic coordinates (latitude, longitude) as input.

The model supports:

- Binary classification (solar present / not present)
- Pixel-level segmentation of detected PV panels
- Area estimation of detected panels
- Explainable outputs via visual overlays

2. Intended Use

Primary Use Case

- Automated detection of rooftop solar PV installations for:
 - Energy planning
 - Policy analysis

- Infrastructure audits
- Sustainability assessments

Intended Users

- Analysts
- Researchers
- Government or utility stakeholders
- Technical evaluators

Out-of-Scope Use

- Real-time drone imagery
 - Ground-level imagery
 - Legal enforcement decisions without human verification
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3. Training Data

The model was trained using a combination of publicly available rooftop solar panel datasets curated from multiple sources.

Data Sources

- Alfred-Weber Institute Solar Dataset
- LSGI547 Solar Panel Dataset
- Solar Panels (Piscinas & Tennis Table) Dataset

Data Characteristics

- Satellite imagery
- Urban and semi-urban rooftops

- Varying resolutions and lighting conditions
- Diverse rooftop geometries

Labels

- Pixel-level segmentation masks for solar panels
 - Binary presence labels derived from segmentation
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4. Data Preprocessing

- Images resized to a fixed input resolution
 - Annotation normalization across datasets
 - Class unification into a single class: `solar_panel`
 - Dataset merging and balancing
 - Data augmentation (flip, scale, mosaic) during training
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5. Model Training

Training Configuration

- Framework: Ultralytics YOLOv8
- Task: Segmentation
- Image size: 1024 × 1024
- Optimizer: Adam (auto-configured)
- Epochs: 30
- Hardware: NVIDIA Tesla T4 (Kaggle GPU)

Key Metrics (Validation)

- mAP@50 (Segmentation): ~0.80
 - mAP@50–95 (Segmentation): ~0.51
 - Precision (Segmentation): ~0.76
 - Recall (Segmentation): ~0.73
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6. Inference Logic

Multi-Zoom Strategy

To handle scale mismatch between training imagery and real-world satellite images, a multi-zoom inference strategy is applied:

- Multiple zoom levels are extracted around the coordinate
- The model is run on each zoom
- The highest-confidence valid detection is selected

Buffer Logic

- Initial inference within a 1200 sq. ft buffer
 - If no confident detection, fallback to 2400 sq. ft buffer
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7. Output & Explainability

Outputs

- JSON prediction per site
- Overlay images with segmentation masks or bounding boxes

Explainability Artifacts

- Visual overlays for human audit
- Explicit QC status per prediction
- Encoded masks or bounding boxes in JSON

8. Quality Control (QC)

The model assigns one of the following QC statuses:

QC Status	Meaning
VERIFIABLE	Clear visual evidence (solar present or confidently absent)
NOT_VERIFIABLE	Insufficient evidence due to image quality issues

9. Known Limitations

- Performance degrades under:
 - Heavy cloud cover
 - Strong rooftop shadows
 - Low-resolution or stale imagery
 - Small rooftop installations may be missed at coarse zoom levels
 - Model performance depends on imagery quality from external providers
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10. Bias & Risk Considerations

- Bias toward urban rooftop styles present in training data
 - Rural or unconventional rooftop designs may be underrepresented
 - Geographic bias may exist due to dataset source regions
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11. Failure Modes

- False negatives for very small installations
 - NOT_VERIFIABLE outputs for occluded rooftops
 - Image fetch failures due to API or network issues
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12. Retraining & Improvement Guidance

To improve performance:

- Add higher-resolution imagery
 - Include region-specific rooftop styles
 - Fine-tune model with recent satellite images
 - Increase training epochs and dataset diversity
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13. Ethical Considerations

- Model outputs should not be used as sole evidence for legal or financial decisions
 - Human verification is recommended for critical use cases
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14. Contact & Ownership

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